```
import os

classes = ["eiffel_tower", "taj_mahal", "golden_gate_bridge"]

base_dir = "/content/landmark_dataset"
os.makedirs(base_dir, exist_ok=True)

for c in classes:
    os.makedirs(os.path.join(base_dir, c), exist_ok=True)

print("Folders created:", os.listdir(base_dir))
```

```
Folders created: ['golden_gate_bridge', 'eiffel_tower', 'taj_mahal']
```

```
from google.colab import files
import shutil

class_name = "eiffel_tower"  # change this each time you upload for a new class

uploaded = files.upload()  # this opens a file picker, choose your 2 images

for filename in uploaded.keys():
    shutil.move(filename, f"/content/landmark_dataset/{class_name}/{filename}")

print(os.listdir(f"/content/landmark_dataset/{class_name}"))
```

Choose Files No file chosen

```
['WhatsApp Image 2026-07-21 at 8.37.51 PM (1).jpeg']
```

```
Start coding or generate with AI.
```

```
from google.colab import files
import shutil

class_name = "golden_gate_bridge"  # change this each time you upload for a new class

uploaded = files.upload()  # this opens a file picker, choose your 2 images

for filename in uploaded.keys():
    shutil.move(filename, f"/content/landmark_dataset/{class_name}/{filename}")

print(os.listdir(f"/content/landmark_dataset/{class_name}"))
```

Choose Files WhatsApp I… PM (2).jpeg

WhatsApp Image 2026-07-21 at 8.37.51 PM (2).jpeg(image/jpeg) - 63600 bytes, last modified: 7/21/2026 - 100% done

```
Saving WhatsApp Image 2026-07-21 at 8.37.51 PM (2).jpeg to WhatsApp Image 2026-07-21 at 8.37.51 PM (2).jpeg
['WhatsApp Image 2026-07-21 at 8.37.51 PM (2).jpeg']
```

Start coding or generate with AI.

```
from google.colab import files
import shutil

class_name = "taj_mahal"  # change this each time you upload for a new class

uploaded = files.upload()  # this opens a file picker, choose your 2 images

for filename in uploaded.keys():
    shutil.move(filename, f"/content/landmark_dataset/{class_name}/{filename}")

print(os.listdir(f"/content/landmark_dataset/{class_name}"))
```

Choose Files WhatsApp I... PM (3).jpeg

WhatsApp Image 2026-07-21 at 8.37.51 PM (3).jpeg(image/jpeg) - 98751 bytes, last modified: 7/21/2026 - 100% done

```
Saving WhatsApp Image 2026-07-21 at 8.37.51 PM (3).jpeg to WhatsApp Image 2026-07-21 at 8.37.51 PM (3).jpeg
['WhatsApp Image 2026-07-21 at 8.37.51 PM (3).jpeg']
```

```
for c in classes:
    path = os.path.join(base_dir, c)
    print(c, "->", os.listdir(path))
```

```
eiffel_tower -> ['WhatsApp Image 2026-07-21 at 8.37.51 PM (1).jpeg']
taj_mahal -> ['WhatsApp Image 2026-07-21 at 8.37.51 PM (3).jpeg']
golden_gate_bridge -> ['WhatsApp Image 2026-07-21 at 8.37.51 PM (2).jpeg']
```

```
import tensorflow as tf

IMG_SIZE = (224, 224)
BATCH_SIZE = 2

full_ds = tf.keras.utils.image_dataset_from_directory(
    "/content/landmark_dataset",
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    shuffle=True
)

class_names = full_ds.class_names
print("Classes found:", class_names)

# Normalize pixels
normalization_layer = tf.keras.layers.Rescaling(1./255)
full_ds = full_ds.map(lambda x, y: (normalization_layer(x), y))
```

```
Found 3 files belonging to 3 classes.
Classes found: ['eiffel_tower', 'golden_gate_bridge', 'taj_mahal']
```

```
from tensorflow.keras import layers, models

base_model = tf.keras.applications.MobileNetV2(
    input_shape=IMG_SIZE + (3,),
    include_top=False,
    weights='imagenet'
)
base_model.trainable = False

model = models.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dense(128, activation='relu'),
    layers.Dropout(0.3),
    layers.Dense(len(class_names), activation='softmax')
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h5
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
history = model.fit(full_ds, epochs=3)
```

```
Epoch 1/3
2/2 ━━━━━━━━━━━━━━━━━━━━ 5s 77ms/step - accuracy: 0.6667 - loss: 0.8053
Epoch 2/3
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 64ms/step - accuracy: 1.0000 - loss: 0.2943
Epoch 3/3
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 62ms/step - accuracy: 1.0000 - loss: 0.2551
```

```
import numpy as np
from tensorflow.keras.preprocessing import image

# Upload a test image
from google.colab import files
test_upload = files.upload()
test_filename = list(test_upload.keys())[0]

img = image.load_img(test_filename, target_size=IMG_SIZE)
img_array = image.img_to_array(img) / 255.0
img_array = np.expand_dims(img_array, axis=0)

pred = model.predict(img_array)
predicted_class = class_names[np.argmax(pred)]
print("Predicted landmark:", predicted_class)
```

```
Choose Files WhatsApp I… PM (1).jpeg
WhatsApp Image 2026-07-21 at 8.37.51 PM (1).jpeg(image/jpeg) - 62332 bytes, last modified: 7/21/2026 - 100% done
Saving WhatsApp Image 2026-07-21 at 8.37.51 PM (1).jpeg to WhatsApp Image 2026-07-21 at 8.37.51 PM (1).jpeg
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 1s/step
Predicted landmark: eiffel_tower
```